DATE :13/10/2026 **ITA 0329 -MOBILE COMPUTING** **M.SAI CHARAN**

CO 1 AT 2-Short Answers TEST **192472233**

1 . spectrum auction = 45MHz

Company policy sets =25kHz

How many full duplex channels =45*100/2*25=45000/50=900

3) total rools = 100

Each carring 8 time slots for voice

N=Number of *carriers *slot per carrier =100*8=800

4) Base station power =80

Hand set ransmit=4

10 log 10(80/4)

$=10\log_{10}(20)=10(1.3010)=13.01dB$

5) test capture signal =-60dBm

Co-channel interface = -92dBm

Carrier to interface ration (C/I) = -60-(-92)=-60+92=32

6) total bandwidth =42MHz

Channel Bandwidth=300khz

Cluster size =7

$N_{total}=42*1000/300=140$

$N_{cell}=140/7=20$

7) intial received power =64W

Distance increases by a factor of 2

Path iosses exponent /(n=4)

$P2 =64(1/2)^4=64/16=4$

8) Transmit power for double cell radius

Formula: Power factor = $(R_2/R_1)^n$

Substitution: $= (2/1)^2 = 4$

9) Total cluster channels

Formula: Total channels = Cluster size × Channels per cell

Substitution: $= 4 \times 15$

60

10) Doppler shift

Formula: $f_d = f_c v/c$

Convert speed: $100 \times 1000/3600 = 27.78$ m/s

Substitution: $= (900 \times 10^6 \times 27.78)/(3 \times 10^8)$

83.33 Hz

11) Duration of one time slot

Formula: Slot duration = Frame duration ÷ Number of slots

Substitution: $= 5/8$

0.625 ms

12) Number of carrier frequencies

Formula: Number of carriers = Total bandwidth ÷ Carrier spacing

Substitution: $= 24$ MHz$/0.2$ MHz

120 carriers

13) Received power in dBm

Formula 1: $P_{\text{dBm}} = 10\log_{10}(P_{\text{W}} \times 1000)$

Formula 2: Received power = Transmit power − Path loss

Substitution:$P_{\text{dBm}} = 10\log_{10}(8 \times 1000) = 39.03$ dBm

$P_r = 39.03 - 33$= 6.03dBm

15) Cluster call capacity

Formula: Total capacity = Number of cells × Calls per cell

Substitution: $= 6 \times 250$ = 1,500 calls

16) Number of active calls

Formula: Active calls = Calls per second × Average call duration in seconds

Substitution: $= 4000 \times (3 \times 60)$

720,000 calls

17) Total bandwidth

Formula: Total bandwidth = Number of channels × Bandwidth per

Substitution: $= 150 \times 25$ kHz $= 3750$ kHz

3.75 MHz

18 Total guard-band bandwidth

Formula: Total guard bandwidth = (Number of bands − 1) × Guard bandwidth

Substitution: $= (30 - 1) \times 40$ kHz $= 1160$ kHz

1.16 MHz

19) Number of handoffs

Formula: Handoffs ≈ Distance traveled ÷ Cell diameter

Substitution:

Distance traveled = $90 \times 1 = 90$ km

Cell diameter = $2 \times 1.5 = 3$ km

Handoffs = $90/3$

Approximately 30 handoffs

20) Formula: Total users = Regional centers × Base stations per center × Users per

Substitution: $= 8 \times 40 \times 250$

= 80,000 users

21) Formula: Number of cells = Total area ÷ Area per cell

Substitution:

Area per cell = $R^2 = 400^2 = 160{,}000$ m^2 $= 0.16$ km^2

Number of cells = $400/0.16$

2,500 cells

22) Formula: Number of channels = Total spectrum ÷ Bandwidth per channel

Substitution: $= 6000/30$

200 channels

24) Formula: Users = Total bandwidth ÷ Bandwidth per user

Substitution: $= 40$ MHz$/0.4$ MHz

100 users

25) Formula: Total users = Number of base stations × Users per station

Substitution: $= 1200 \times 600$

720,000 users

26) Formula: Paging rounds = Total cells ÷ Cells per round

Substitution: $= 200/4$

50 rounds

28) Roaming subscribers = Total subscribers × Roaming percentage ÷ 100

Substitution: $= 600{,}000 \times 15/100$

90,000 subscribers

29) Formula: Utilization = (Current load ÷ Maximum capacity) × 100

Substitution: $= 9000/12000 \times 100$

75%

30) Formula: Channels per cell = Total bandwidth ÷ Reuse factor

Substitution: $= 48{,}000/200/4$

60 channels